

SIMRAN KARAMCHANDANI

U.S. Citizen

svkaramchandani@gmail.com | (201) 786-3665

<https://linkedin.com/in/skaramchandani> | <https://github.com/simkaramchandani01>

OBJECTIVE

Data Analytics Engineer with experience in workflow optimization, analytics, development of AI-powered applications, and collaboration with GenAI tools. Seeking AI Engineer roles to design data-driven and intelligent business solutions.

SKILLS & SYSTEMS

- Data Analytics: **Databricks**, ETL, Data Cleansing, Exploratory Data Analysis, Data Mining, Data Modeling
- Systems: **n8n Automation**, **Agentic AI**, CrewAI, Tableau, **Power BI**, **Power Apps**, Veeva Vault QMS, LIMS, MES
- Programming: **SQL**, **Anthropic Claude**, **Python**, R, **LLM**, MATLAB/Simulink, Linux Bash, Java
- Project Management: **Jira**, **Confluence**, Yellow Belt Six Sigma & Green Belt Six Sigma Certified, Agile Scrum
- Machine Learning Algorithms: Regression, Classification, Clustering, Gradient Boosting, XGBoost, LightGBM

WORK EXPERIENCE

Takeda Pharmaceuticals, *Lexington, MA*

Jul 2025 – Present

Data Visualization & Analytics Engineer, Data Digital & Technology (DD&T) Biologics

- Developed and deployed **Power BI** dashboards across 5+ functions, supported by automated **ETL** workflows in **Databricks**, reducing manual data handling by 40% and enabling real-time site performance visibility for global and site leadership.
- Led the **Digital Community of Practice** by organizing and presenting 9 lunch-and-learn presentations to the site to strengthen digital maturity and promote AI awareness and engagement.

Vertex Pharmaceuticals, *Boston, MA*

Jan 2022 – Jul 2025

External Manufacturing Operations Senior Specialist, Type I Diabetes (T1D) Cell Therapy

May 2024 – Jul 2025

- Developed an interactive **Power BI** dashboard to track commercial manufacturing batch disposition TAT across contract organizations (CDMOs), identifying slowest performing assays and record completions.
- Resolved 95% of CDMO production issues within target timelines through cross-functional quality management.

Manufacturing Operations Specialist (Shift Lead), Type I Diabetes (T1D) Cell Therapy

Feb 2023 – May 2024

- Improved Manufacturing efficiency by 30% through delivery of a **Power BI** dashboard tracking shift KPIs.
- Led shift operations, delegated personnel responsibilities, and acted as SME for process operations and event triage.

Manufacturing Operations Senior Scientific Associate, Type I Diabetes (T1D) Cell Therapy

Jan 2022 – Feb 2023

- Supported tech transfer to new facility via material readiness and execution of qualification runs.

EDUCATION

- **University of Chicago**, M.S. Applied Data Science, **GPA: 3.97/4.0** Jul 2023 – Jun 2025
- **Northeastern University**, M.S. Chemical Engineering, **GPA: 3.7/4.0** Aug 2021– Dec 2022
- **Syracuse University**, B.S. Bioengineering and Neuroscience, **GPA: 3.5/4.0** Aug 2017– May 2021

PROJECTS

Independent Projects

CRAVINGS – AI -Powered Food Discovery App (**Lovable AI**):

- Developed **natural language-driven** application that recommends recipes or nearby restaurants based on user cravings and self-reported cooking skill level.
- Integrated community features, allowing users to share recipes and engage in food-specific interest groups, enhancing personalization and user engagement.

Email Summarization Automation (**n8n**):

- Built an automated workflow using n8n to generate daily summaries of unread emails, including: sender, subject, content highlights, and direct email links.
- Implemented **AI-driven** prioritization to rank emails by urgency, improving inbox efficiency.

University of Chicago

Jul 2023 – Jun 2025

Lease-to-Own Pricing Strategy (Capstone): Won “Best in Show” in Winter 2025 Capstone Showcase

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- Built a **predictive model** using customer behavior and financial patterns to identify which approved applicants were most likely to sign a lease, driving **data-informed pricing strategies**.

Generative AI:

- Developed multimodal chatbot that returns product information and displays relevant product images based on user queries, leveraging text and image processing with **TorchVision**, **CLIP embeddings**, and Facebook AI Similarity Search (**FAISS**) for similarity search and retrieval from vector storage.
- Built **CrewAI** resume tailoring tool to parse job descriptions and auto-tailor resumes to role requirements.

Predictive Modeling & Data Analytics:

- Developed Machine Learning (ML) models in **Python** to assess Type II Diabetes risk, analyze audio features for music success, and predict gym membership churn based on user behavior.